Sharan Giri

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Professional Statement

2 years shipping production automation pipelines at Barclays for Markets and Private Banking under SLA-bound delivery and live production support. Extending that foundation into end-to-end LLM and RAG systems including multi-turn agentic pipelines, hybrid retrieval, and MLflow-instrumented observability across latency, cost, and drift, with deployments on GCP and Kubernetes.

Education

Khoury College of Computer Science, Northeastern University — Boston, MA
M.S. in Data Science (GPA: 4.0) — Sep 2024 – May 2026
- Secured Third Place at Google-Northeastern Expo for AudioSeek (Dec 2025)

Sri Sivasubramaniya Nadar College of Engineering, Anna University — India
B.E. in Computer Science Engineering (GPA: 3.7) — Aug 2018 – Jun 2022

EXPERIENCE

Barclays Global Service Centre – Markets And Private Banking Technology — Aug 2022 – Jun 2024
Software & Automation Engineer | Data Pipeline and API Integration — India
- Built **production-grade pipelines** processing **ML extraction outputs** from unstructured email and API/JSON sources into structured formats, improving throughput by **50%** and saving **85 hours/month**
- Enforced **error volume thresholds** and automated stakeholder alerting across **6 production workflows**, reducing false-positive alerts by **72%** and improving system reliability
- Migrated **4** UI-based automations to **API-driven components**, cutting turnaround by **33%** and improving **reliability**
- Owned **SLA-bound production support** with a **2-day** resolution target, debugging live failures, coordinating **controlled rollbacks** and redeploying after regression testing

Projects

CourseLENS | Governed Multi-Turn AI Reasoning System | *SIGDIAL 2026 (Under Review)* — Jan 2026 – Apr 2026
- Engineered an **end-to-end ingestion pipeline** transforming raw PPTX and PDF assets into a **parent-child vector store** with curriculum-gated boundaries to scope retrieval by week
- Architected **4-stage** policy engine with deterministic routing and immutable logs, achieving **100%** traceable LLM outputs
- Built **90/10 hybrid retrieval** as an **anti-hallucination boundary**, confining LLM reasoning to **top-5 verified chunks**
- Validated **100% policy compliance** via adversarial stress testing and achieved **96.15%** Ragas-scored data faithfulness

AudioSEEK | Production RAG Platform | *Google-Northeastern Expo* — Sep 2025 – Dec 2025
- Architected a **production RAG platform** with **admin dashboard** for real-time ingestion status, converting **30**+ hours of audio into a searchable knowledge base with answers under **15 seconds**
- Instrumented **monitored ingestion pipeline** with async transcription, restart controls, and completion notifications
- Engineered **hybrid SQL-FAISS** store with temporal constraints to prevent **lookahead bias** and enforce data integrity
- Deployed **Dockerized** GKE microservices via **Terraform** with **MLflow** for latency tracking and production drift detection

FinAgent | Document Intelligence and Retrieval Platform — Mar 2026 – Apr 2026
- Architected a **two-stage** pipeline blending NLP extraction with generative reasoning across **72** SEC 10-K filings
- Built **multi-threaded ingestion** engine parallelizing numerical and textual filing data, cutting **LLM token costs** by **98%**
- Developed a **normalization layer** deduplicating XBRL tags and aligning **5-year time-series** across **16** companies, with **FAISS** retrieval over **BGE-M3** embeddings and **FastAPI**-exposed endpoints reducing latency to **sub-90s**
- Enforced **Pydantic schema validation** across all LLM outputs, achieving **95%**+ faithfulness on **1,500**+ structured signals

Early Product Success Prediction | Demand Signal Modeling and Feature Engineering — Nov 2025 – Dec 2025
- Built a **PySpark ETL pipeline** transforming raw review data into a model-ready dataset for **28-day launch prediction**
- Engineered **40 structured signals** across rating trends, sentiment polarity, and review velocity under **A/B ablation**
- Trained **XGBoost** models across multiple signal modalities achieving **Lift@100** of **2.6x** over baseline and **Precision@100** of **0.65**, outperforming all single-modality models
- Delivered **SHAP-backed explainability** over final predictions enabling interpretable and **auditable** signal attribution

Technical Skills

Software & Data Engineering: Python, SQL, REST APIs, FastAPI, PySpark, Pandas, Airflow, Git, CI/CD (GitHub Actions)
AI Systems & Retrieval: RAG Pipelines (Retrieval Augmented Generation), Agentic AI Systems, Multi-Agent Orchestration, LangChain, LangGraph, FAISS, Vector Databases, BM25, BGE-M3, HuggingFace, Pydantic, Prompt Engineering, Structured Output Generation
ML & Modeling: PyTorch, Scikit-learn, XGBoost, Feature Engineering, SHAP, Ragas, LLM Evaluation, ML Observability
MLOps & Infrastructure: Docker, Kubernetes (GKE), MLflow, Terraform
Cloud Platforms: GCP (Cloud Run, GCS, Firestore, Cloud SQL), Gemini Flash